Romit Chakraborty

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Profile

Quantum chemist and architect of phase-space electronic structure theory. My research develops Quantum Boltzmann Equation (QBE) solvers that regularize mean-field singularities and ameliorate the exponential scaling walls of strongly correlated quantum matter. Building on postdoctoral research at UC Berkeley and doctoral training at the University of Chicago, I now direct Point Reyes Sound, Inc., translating thermodynamic theory into scalable, hardware-agnostic architectures for next-generation quantum simulation.

Skills

2023--now	**ML Infrastructure**	PyTorch, JAX, CUDA, Multi-GPU, LLMs, LoRA
2023--now	**Quantum AI & Fault Tolerance**	FTQC Architecture, Quantum Error Correction (QEC), PySCF
2024--now	**Data Pipelines & Synthetic Data**	Synthetic Data Generation, Vector Databases, ChromaDB, Azure
2012--now	**HPC & Systems Engineering**	Python, C++, MPI, Slurm, GPU Acceleration, Linux
2012--now	**Quantum Simulation**	Quantum Boltzmann Equation (QBE), Q-Chem

Experience

June 2026	**Founder & CEO**	Point Reyes Sound, Inc., San Francisco, CA
2024-2025	**Senior Computational Scientist, Quantum Solutions**	PsiQuantum, Palo Alto, CA, USA
2023-2024	**Staff Researcher in Quantum AI**	University of Chicago, Chicago, IL, USA
2017-2023	**Postdoctoral Research Fellow**	UC Berkeley & LBNL, Berkeley, CA, USA

Education

University of California, Berkeley and Lawrence Berkeley National Laboratory — *Berkeley, CA, USA*
Postdoctoral Research Fellow in Computational Materials Science — *2017-2023*
Advisor: Prof. Martin Head-Gordon

University of Chicago — *Chicago, IL, USA*
MS, Ph.D. in Theoretical and Computational Quantum Chemistry — *2012-2017*
Advisor: Prof. David A. Mazziotti

IIT Bombay — *Bombay, India*
Master of Science — *2009-2012*

St Stephen's College, Delhi — *Delhi, India*
B.Sc (Hons) in Chemistry, with Physics and Mathematics — *2006-2009*

Appointments

Point Reyes Sound, Inc. *San Francisco, CA*

Founder & CEO *June 2026 -- Present*

- **Phase-Space Electronic Structure:** Formulated a Hessian-free orbital optimization framework utilizing continuous phase-space fluid advection and the Bhatnagar-Gross-Krook (BGK) collision operator to autonomously regularize mean-field singularities.
- **Quantum Boltzmann Entrainment Tensor (QBET):** Developed a theoretical methodology to restrict quantum simulation workloads strictly to the active space, effectively resolving exponential scaling limits in strongly correlated multireference systems.
- **Solid-State k-Space Decoupling:** Generalized the phase-space transport engine for periodic crystalline systems by projecting continuous Wigner distributions onto cGTOs, mapping multidimensional transport to decoupled matrix relaxations across the Brillouin zone.

PsiQuantum, Corp. *Palo Alto, CA*

Senior Computational Scientist *Oct 2024 -- Mar 2025*

- Translated quantum chemical models of correlated electron systems into robust algorithmic frameworks with scalable software architectures.

University of Chicago *Chicago, IL*

Staff Researcher *Nov 2023 -- Oct 2024*

- **Physics-Grounded Generative Workflows:** Led the development of a symmetry-aware framework conditioning generative models on strict physical constraints (molecular geometry, charge/spin manifolds, point-group symmetry) to resolve out-of-distribution mode collapse.
- **Physical Symmetry Integration:** Pioneered the integration of physical symmetries into large-scale generative architectures, securing Microsoft Azure Foundation Models Research funding to bridge theoretical chemistry and machine learning.
- **Mathematical Verification Metrics:** Engineered rigorous protocols for generating mathematically-verified synthetic data, establishing evaluation metrics defined by exact structural equivalences rather than heuristic approximations.

Publications

[+] *Equal Contributors*

Technical Reports

Rendering the Quantum Fluid: Mapping Phase-Space Electronic Structure to Tile-Based Deferred Shading, *Point Reyes Sound Technical Whitepaper PRS-TR-2026-02,* ***2026***. [PDF] [Interactive Dossier]

The Quantum Boltzmann Entrainment Tensor: A Hessian-Free Framework for Multireference Orbital Optimization, *Point Reyes Sound Technical Report PRS-TR-2026-01 / In Preparation for JCTC,* ***2026***. [PDF] [Interactive Dossier]

Quantum Resource Estimation: Hardware Resource Compilation for Active-Space Driven Fault-Tolerant Quantum Simulation, *Point Reyes Sound Technical Whitepaper PRS-TR-2026-03,* ***2026***. [PDF]

Peer Reviewed Articles and Preprints

Chakraborty, R. ***2026***. **Quantum Boltzmann Equation Self-Consistent-Field for the Entropic Regularization of Mean-Field Singularities**. *arXiv:2608.14979 [quant-ph, physics.chem-ph]*. [arXiv:2608.14979] [PDF]

Chakraborty, R., Talbot, J. J., Shen, H., Yabuuchi, Y., Jiang, H., Carsch, K.M., Furukawa, H., Long, J.R., Head-Gordon, M. ***2024***. Quantum Chemical Modeling of Hydrogen Binding in Metal-Organic Frameworks: Validation, Insight, Predictions and Challenges. *Selected as a Hot Article*. Phys. Chem. Chem. Phys. 26 (8), 6490-6511

Carsch, K.M.,**Chakraborty, R.**, Head-Gordon, M, Long, J.R. et. al. . ***2023***. Selective Adsorption of Oxygen from Humid Air in a Metal–Organic Framework with Trigonal Pyramidal Copper(I) Sites. J. Am. Chem. Soc. 2024, 146, 5, 3160–3170

Talbot J.J., **Chakraborty, R.**, Shen, H., Head-Gordon, M. ***2023***. A Free Energy Decomposition Analysis: Insight into Binding Thermodynamics from Absolutely Localized Molecular Orbitals. *J. Phys. Chem. Lett.*, 14, 5432-5440.

Funke L.M., **Chakraborty R.**, Carsch, K.M., Head-Gordon, M., Long, J.R., Reimer, J.A..***2023***. Assessment of Adsorbate π-backbonding in Cu(I) Metal–Organic Frameworks via Multinuclear NMR Spectroscopy and Density Functional Theory Calculations. *J. Phys. Chem. C* 127 (15): 7513–7519

Chakraborty, R., Carsch, K.M., Jaramillo, David E, Yabuuchi Tuto, Furukawa H, Long, J.R., Head-Gordon M. ***2022***. Prediction of Multiple Hydrogen Ligation at a Vanadium(II) Site in a Metal-Organic Framework. *J. Phys. Chem. Lett.*, 13(44): 10471-10478.

Barnett, B.R., Evans, H.A, Su, G.M., Jiang, H.Z.H., **Chakraborty, R.**, Banyeretse, D., Hartman, T.J., Martinez, M.B., Trump, B.A., Tarver, J.D., Dods, M.N., Funke, L.M., Borgel, J., Reimer, J.A., Drisdell, W.A., Hurst, K.E., Gennett, T., FitzGerald, S.A., Brown, C.M., Head-Gordon, M., Long, J.R. ***2021***. Observation of an Intermediate to H2 Binding in a Metal–Organic Framework. 2021. *J. Am. Chem. Soc.*, 143(36): 14884-14894.

Epifanovsky E., Gilbert A.T.B., Feng X., Lee J., Mao Y., Mardirossian N., ..., **Chakraborty, R.**, ..., Head-Gordon, M., Herbert J.M., Krylov, A.I. Software for the frontiers of quantum chemistry: An overview of developments in the Q-Chem 5 package. ***2021***. *J. Chem. Phys.*, 155(8):084801

Jaramillo, D.E., Jiang, H.Z.H., Evans, H.A., **Chakraborty, R.**, Furukawa, H., Brown, C.M., Head-Gordon, M., Long, J.R. ***2021***. Ambient-temperature hydrogen storage via vanadium (II)-dihydrogen complexation in a metal–organic framework. *J. Am. Chem. Soc.*, 143(16):6248:6256

Stauch T.$^{+}$, **Chakraborty, R.**$^{+}$, Head-Gordon, M. 2019. Quantum Chemical Modeling of Pressure Induced Spin Crossover in Octahedral Metal Ligand Complexes. ***2019***. *ChemPhysChem*, 20(21):2742-2747

Chakraborty, R., Mazziotti, D.A. ***2018***. Sparsity of the wavefunction from the generalized Pauli exclusion principle. *J. Chem. Phys.* 148 (5), 054106.

Chakraborty, R., Mazziotti, D.A. 2017. Noise-assisted energy transfer from the dilation of the set of one electron reduced density matrices. ***2017***. *J. Chem. Phys.* 146 (18), 184101.

Chakraborty, R., Mazziotti, D.A. ***2016***. Role of the generalized Pauli constraints in the quantum chemistry of excited states. *Int. J. Quantum Chem.* 116 (10), 784-790.

Chakraborty, R., Mazziotti, D.A. ***2015***. Structure of the one electron reduced density matrix from the generalized Pauli exclusion principle. *Int. J. Quantum Chem.* 115 (19), 1305-1310.

Chakraborty, R., Mazziotti, D.A. ***2015***. Sufficient condition for the openness of a many electron quantum system from the violation of a generalized Pauli exclusion principle. *Physical Review A* 91 (1), 010101

Datta, S. N., Ghosh A., **Chakraborty, R.**. ***2015***. Variational aspects of the Klein–Gordon equation. *Ind. J. Phys.* 89 (2), 181-187.

Chakraborty, R., Mazziotti, D.A. ***2014***. Generalized Pauli conditions on the spectra of one electron reduced density matrices of atoms and molecules. *Phy. Rev. A* 89 (4), 042505.

`https://scholar.google.com/citations?hl=en&user=m4HlFRIAAAAJ&view_op=list_works&sortby=pubdate`

Awards

Fall, 2023	**Accelerate Foundation Models Research**, Microsoft Azure
Smr, 2023	**Award for Contributions to Inclusion in Science and Society**, Berkeley Lab (Postdoc Assoc.)
Smr, 2023	**Successful EOI for Independent Early Career Fellowships**, University of Oxford, UK
Fall, 2015	**Arts Science and Culture Graduate Fellowship**, University of Chicago, USA
Smr, 2009	**Lindau Graduate Fellowship**, German Research Foundation (DFG)

Professional Development

Peer Review

Nature Communications, 2022
Journal of Chemical Theory and Computation, 2023
Molecular Physics, 2018, 2019, 2020
Journal of Physics: Condensed Matter, 2017

Professional Memberships

American Physical Society
Councillor (Alternate) at California Section of the American Chemical Society (2023-2026)
Program Book Coordinator at the Western Regional Meeting (October 2025)
Project SEED (Summer Opportunities for Economically Disadvantaged Students) (2025-2026)
Member at Large, Cal ACS 2026-2027